Process dynamics and control

A refresher from the very start, with the rocking-bag controller as the running example

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Companion to the closed-loop cell therapy repository

How to read this

This covers the core of an undergraduate course in process dynamics and control. It assumes only high-school calculus: derivatives, integrals and the exponential function. Each idea is introduced with a simple tank or the bag's dissolved oxygen, then used again later. Part I is about how processes respond on their own (*dynamics*). Part II is about how a controller changes that (*feedback*). Part III covers the practical issues every real controller has, and ends with the two loops in the rocker. Exercises with answers are at the end.

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1 What control is

1.1 An everyday loop

You step into a shower that is too cold. You feel the water (*measure*), compare it with what you want (*setpoint*), and turn the hot tap (*act*). You wait, feel again and adjust again. That is feedback control. Three things make it harder than it sounds:

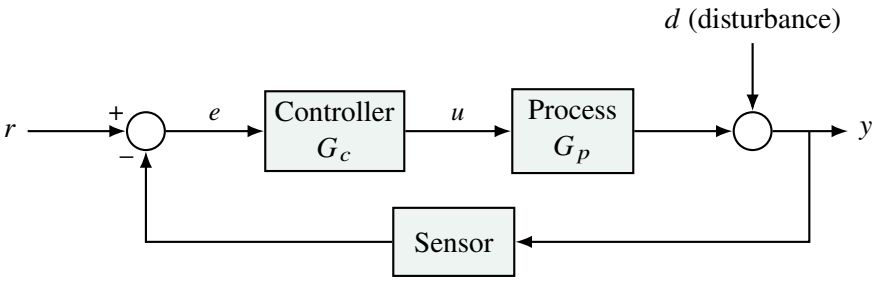
- **Delay.** Hot water takes a few seconds to reach you. If you keep turning the tap during that time, you scald yourself, then over-correct and freeze. Delay is the main enemy of feedback.
- **Disturbances.** Someone flushes a toilet and the cold pressure drops. You did not change anything, but the temperature moved.
- **Limits.** The tap only turns so far.

1.2 Vocabulary

Term	Symbol	In the shower / in the bag's DO loop
Controlled variable (process variable)	y	water temperature / dissolved oxygen
Setpoint	r	the temperature you want / 50% air saturation
Error	$e = r - y$	too cold by 5 °C / 10% below setpoint
Manipulated variable	u	hot tap position / rock rate, then O_2 in the gas
Disturbance	d	toilet flush / cells using more oxygen as they grow
Sensor		your skin / optical DO patch
Actuator		the tap / the rocker motor and gas valves
Controller	G_c	you / the PI algorithm in the rocker's firmware

1.3 The block diagram

Control engineers draw every loop the same way. Signals travel along arrows; boxes change signals; circles add or subtract them.



The minus sign at the left circle is what makes it *negative* feedback: if y rises above r, the error goes negative and the controller pushes back.

Part I

Process dynamics: how things respond on their own

2 Models come from balances

Every model in this field starts from the same sentence:

$$\text{rate of accumulation} = \text{rate in} - \text{rate out} + \text{rate of generation}. \tag{1}$$

Example 1: a tank draining through a valve

A tank of cross-section A is filled at flow q_{in} and drains through a valve. For a simple valve, outflow is proportional to level: $q_{\text{out}} = h/R$, where R is the valve's resistance. A volume balance gives

$$A\,\frac{\mathrm{d}h}{\mathrm{d}t} = q_{\text{in}} - \frac{h}{R}.$$

Example 2: dissolved oxygen in the rocking bag

Oxygen enters through the liquid surface at a rate proportional to how far the liquid is from saturation, and the cells consume it at rate OUR:

$$\frac{\mathrm{d}O}{\mathrm{d}t} = k_L a\,(C^* - O) - \text{OUR}.$$

$k_L a$ is set by the rock rate; C^* by the oxygen fraction in the gas.

Steady state means nothing changes: set the derivative to zero. For the tank, $\bar{h} = R\,\bar{q}_{\text{in}}$. For the bag, $\bar{O} = C^* - \text{OUR}/k_L a$. The bar marks a steady-state value.

3 Deviation variables and linearisation

3.1 Deviation variables

Control is about *changes* around an operating point, so we measure everything from steady state:

$$y'(t) = y(t) - \bar{y}, \qquad u'(t) = u(t) - \bar{u}.$$

At the starting steady state every deviation variable is zero. That makes the maths much cleaner, as you will see with the Laplace transform. From here on the primes are usually dropped: y and u mean deviations unless stated otherwise.

3.2 Linearisation

Most real processes are nonlinear. Near an operating point, any smooth function looks like a straight line (first-order Taylor expansion):

$$f(x) \approx f(\bar{x}) + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{\bar{x}} (x - \bar{x}). \tag{2}$$

Example: a real valve

A real valve drains as $q_{\text{out}} = \beta\sqrt{h}$, which is nonlinear. Around $\bar{h}$,

$$\beta\sqrt{h} \approx \beta\sqrt{\bar{h}} + \frac{\beta}{2\sqrt{\bar{h}}}\,(h - \bar{h}),$$

so near $\bar{h}$ the valve behaves like a linear one with resistance $R = 2\sqrt{\bar{h}}/\beta$. The price: the model is only accurate for small changes, and R depends on where you operate.

Linear models are what all the tools below need. In the bag, pH depends on $\log_{10}(y_{\text{CO}_2})$; section 18 linearises it in exactly this way.

4 First-order systems

4.1 The standard form

Rearrange the tank balance by multiplying through by R:

$$\underbrace{AR}_{\tau}\,\frac{\mathrm{d}h}{\mathrm{d}t} + h = \underbrace{R}_{K}\,q_{\text{in}}.$$

Any process that can be written as

$$\tau\,\frac{\mathrm{d}y}{\mathrm{d}t} + y = K\,u \tag{3}$$

is called **first order**. Only two numbers describe it:

- the **gain** K: how much y finally changes per unit change in u;
- the **time constant** τ: how fast it gets there.

The bag's oxygen balance has the same form (divide by $k_L a$): $\tau = 1/k_L a$, which is 0.3 to 1 hour in the rocker.

4.2 The step response, solved by hand

At $t = 0$ the input jumps by M and stays there (a *step*). With $y(0) = 0$, separate the variables:

$$\frac{\mathrm{d}y}{KM - y} = \frac{\mathrm{d}t}{\tau} \quad\Longrightarrow\quad -\ln(KM - y) = \frac{t}{\tau} + c \quad\Longrightarrow\quad \boxed{y(t) = KM\left(1 - e^{-t/\tau}\right).}$$

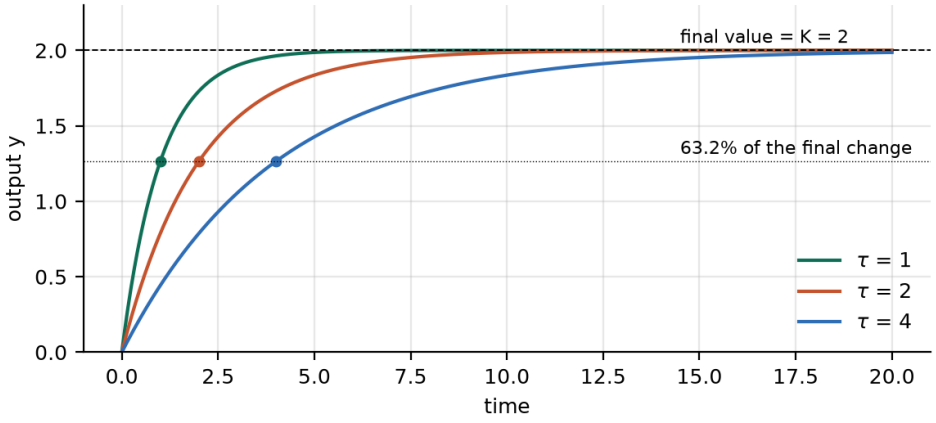


Figure 1: Step responses of a first-order system with $K = 2$, $M = 1$. The dots mark $t = \tau$.

Three facts to remember:

- At $t = \tau$ the response has covered $1 - e^{-1} = 63.2\%$ of its final change. This is how you read τ off a plot.
- After 3τ it is at 95%; after 5τ, 99%. "Five time constants" is the practical settling time.
- The initial slope is KM/τ: a first-order system responds immediately, without an S-shape.

5 The Laplace transform

5.1 Why bother

Solving every ODE by separating variables gets painful once you connect a controller, a process and a sensor together. The Laplace transform turns **differential equations into algebra**: derivatives become multiplication by s, and the solution of a whole loop is found by multiplying and dividing fractions.

5.2 Definition

For a function $f(t)$ that starts at $t = 0$,

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t)\, e^{-st}\, \mathrm{d}t. \tag{4}$$

s is just a new variable (it can be complex). You rarely compute this integral: you use a table. Two entries, worked out once to see where they come from:

$$\mathcal{L}\{1\} = \int_0^\infty e^{-st}\, \mathrm{d}t = \frac{1}{s}, \qquad \mathcal{L}\{e^{-at}\} = \int_0^\infty e^{-(s+a)t}\, \mathrm{d}t = \frac{1}{s+a}.$$

5.3 The rule that makes it useful

Integrating by parts,

$$\mathcal{L}\left\{\frac{\mathrm{d}y}{\mathrm{d}t}\right\} = \Big[y\, e^{-st}\Big]_0^\infty + s\int_0^\infty y\, e^{-st}\, \mathrm{d}t = sY(s) - y(0). \tag{5}$$

With deviation variables, $y(0) = 0$, so **differentiating is multiplying by** s. By the same logic, integrating is dividing by s.

5.4 The table you actually need

Time domain $f(t)$	Laplace domain $F(s)$	Comment
unit step, 1 for $t \geq 0$	$1/s$	a sudden constant change
step of size M	M/s	
e^{-at}	$1/(s+a)$	exponential decay
t	$1/s^2$	a ramp
$\sin\omega t,\ \cos\omega t$	$\omega/(s^2+\omega^2),\ s/(s^2+\omega^2)$	oscillation
$\mathrm{d}y/\mathrm{d}t$	$sY(s) - y(0)$	$= sY(s)$ for deviation variables
$\int_0^t y\, \mathrm{d}t$	$Y(s)/s$	
$y(t-\theta)$, a delay of θ	$e^{-\theta s}\, Y(s)$	time delay
Final value theorem: $\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s)$ (if y settles)		
Initial value theorem: $\lim_{t\to 0^+} y(t) = \lim_{s\to\infty} sY(s)$		

The final value theorem is the workhorse: it tells you where a response ends up *without* solving for $y(t)$.

5.5 Going back: partial fractions

To get $y(t)$ from $Y(s)$, split $Y(s)$ into simple fractions that are in the table. For the first-order step: transform (3) with $u = M/s$:

$$\tau s\, Y + Y = K\,\frac{M}{s} \quad\Longrightarrow\quad Y(s) = \frac{KM}{s(\tau s+1)} = KM\left[\frac{1}{s} - \frac{1}{s+1/\tau}\right] \quad\Longrightarrow\quad y(t) = KM\left(1 - e^{-t/\tau}\right).$$

Same answer as before, with no integration.

6 Transfer functions

6.1 Definition

The **transfer function** of a process is the ratio of the Laplace transform of its output to that of its input, with all deviation variables starting at zero:

$$G(s) = \frac{Y(s)}{U(s)}. \tag{6}$$

It is the process's "personality", independent of what input you apply. Given any input $U(s)$, the output is simply $Y(s) = G(s)\,U(s)$. For the first-order system (3):

$$\boxed{G(s) = \frac{K}{\tau s + 1}} \tag{7}$$

6.2 Reading a transfer function

- **Steady-state gain** = $G(0)$. By the final value theorem, a step of size M ends at $\lim_{s\to 0} s\,G(s)\,M/s = G(0)\,M$.
- **Poles** are the values of s that make the denominator zero. For $K/(\tau s + 1)$ there is one pole at $s = -1/\tau$. Each pole p contributes a term e^{pt} to the response, so poles set the *speed and shape* of the response (next section).
- **Zeros** are the values of s that make the numerator zero. They change how much each pole's term contributes, but not stability.

6.3 A small zoo

Process	$G(s)$	Example
Pure gain	K	pH responding instantly to CO_2 in the gas (section 18)
First order	$K/(\tau s + 1)$	tank with valve; DO in the bag
Integrating	K/s	tank drained by a fixed-speed pump: level keeps rising if inflow exceeds outflow
Time delay	$e^{-\theta s}$	liquid travelling down a pipe to a sensor; an analyser's turnaround
First order plus dead time (FOPDT)	$Ke^{-\theta s}/(\tau s + 1)$	the most common model used for tuning; fits most process step tests
Two in series	$\frac{K_1}{\tau_1 s + 1} \cdot \frac{K_2}{\tau_2 s + 1}$	two tanks, one draining into the next

Processes in series simply **multiply**. That is the main convenience of transfer functions.

7 Poles and stability

Partial fractions always break a response into terms like $c\,e^{pt}$, one per pole p. Write $p = a + ib$:

$$e^{pt} = e^{at}\left(\cos bt + i \sin bt\right).$$

- The **real part** a decides growth or decay: $a < 0$ decays, $a > 0$ grows without limit.
- The **imaginary part** b decides oscillation: $b \neq 0$ means the response oscillates at frequency b. Complex poles always come in conjugate pairs.

A linear system is **stable** if and only if **every pole has a negative real part**, that is, every pole lies in the left half of the complex plane. Poles further left mean faster responses.

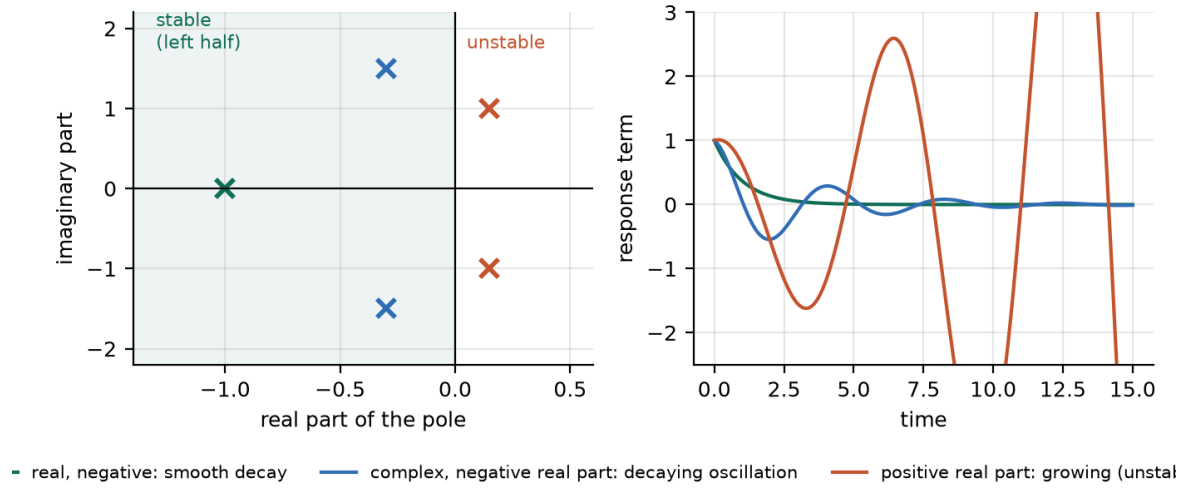


Figure 2: Where a pole sits (left) decides what its term does in time (right).

8 Second-order systems

Two first-order processes in series, or a first-order process under a PI controller, give a second-order system:

$$G(s) = \frac{K}{\tau^2 s^2 + 2\zeta\tau s + 1}. \tag{8}$$

τ sets the speed; the **damping ratio** ζ sets the shape. The poles are $s = \left(-\zeta \pm \sqrt{\zeta^2 - 1}\right)/\tau$:

ζ	Poles	Step response
$\zeta > 1$	two real, negative	overdamped: slow, S-shaped, no overshoot
$\zeta = 1$	one repeated real	critically damped: fastest without overshoot
$0 < \zeta < 1$	complex pair	underdamped: overshoots and oscillates
$\zeta \le 0$	on or right of the imaginary axis	sustained or growing oscillation: unstable

For an underdamped system the overshoot is $\exp\left(-\pi\zeta/\sqrt{1-\zeta^2}\right)$: 53% at $\zeta = 0.2$, 16% at $\zeta = 0.5$. Many controllers are tuned for ζ around 0.7–1.

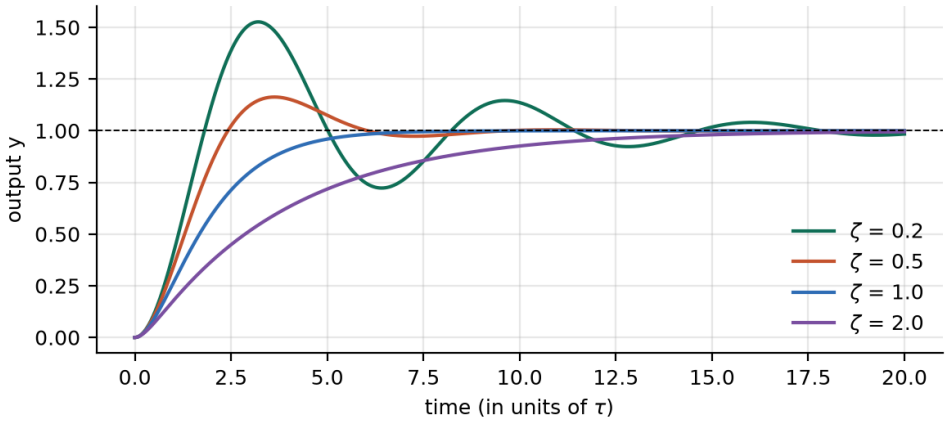


Figure 3: Second-order step responses for several damping ratios ($K = 1$).

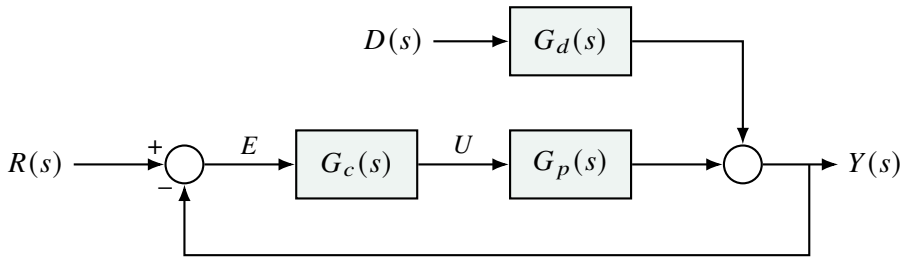
Part II
Feedback control

9 Closing the loop

9.1 Block-diagram algebra

Three rules are enough:

- **Series**: boxes one after another multiply, $G_1 G_2$.
- **Parallel**: signals added at a circle add, $G_1 + G_2$.
- **Feedback**: the loop below.



Read the diagram as equations: $E = R - Y$, $U = G_c E$, $Y = G_p U + G_d D$. Eliminate E and U:

$$Y = G_p G_c (R - Y) + G_d D \quad \Longrightarrow \quad \boxed{Y(s) = \frac{G_c G_p}{1 + G_c G_p} R(s) + \frac{G_d}{1 + G_c G_p} D(s)}$$

The first fraction says how well y follows the setpoint; the second, how well disturbances are rejected. Both share the denominator $1 + G_c G_p$, and

> the **closed-loop poles** are the roots of the **characteristic equation** $1 + G_c(s)\, G_p(s) = 0$. The controller moves the poles. That is all control design really is.

10 The proportional controller and its offset

The simplest controller is proportional: $u = K_c\, e$, so $G_c = K_c$. Apply it to a first-order process $G_p = K/(\tau s + 1)$ and a unit setpoint step $R = 1/s$:

$$Y(s) = \frac{K K_c}{\tau s + 1 + K K_c} \cdot \frac{1}{s} = \frac{K K_c}{1 + K K_c} \cdot \frac{1}{\tau_{\mathrm{cl}}\, s + 1} \cdot \frac{1}{s}, \qquad \tau_{\mathrm{cl}} = \frac{\tau}{1 + K K_c}.$$

Two conclusions:

- The loop is **faster** than the process alone: $\tau_{\mathrm{cl}} < \tau$.
- By the final value theorem, $y(\infty) = K K_c/(1 + K K_c) < 1$. There is always a **steady-state offset**. With $K = 2$ and $K_c = 1$, y settles at $2/3$ of the setpoint (figure 4).

Why, intuitively: to hold y away from zero, the process needs a non-zero u. A P controller only produces non-zero u when there is non-zero error. So some error must remain. Raising K_c shrinks the offset but, as section 12 shows, too much gain destabilises the loop.

11 Integral action: the PI controller

Add a term proportional to the *accumulated* error:

$$u(t) = K_c\left(e(t) + \frac{1}{\tau_I}\int_0^t e\,\mathrm{d}t\right), \qquad G_c(s) = K_c\left(1 + \frac{1}{\tau_I s}\right) = K_c\,\frac{\tau_I s + 1}{\tau_I s}. \tag{9}$$

τ_I is the **integral time** (the code calls it `ti_h`). Smaller τ_I means stronger integral action.

Why the offset disappears. If the loop settles, every signal is constant, including u. But u contains $\int e\,\mathrm{d}t$, which can only be constant if $e = 0$. In transfer-function terms: $G_c(s) \to \infty$ as $s \to 0$, so $G_c G_p/(1 + G_c G_p) \to 1$ at $s = 0$, and by the final value theorem $y(\infty) = r$. This holds for any constant disturbance too.

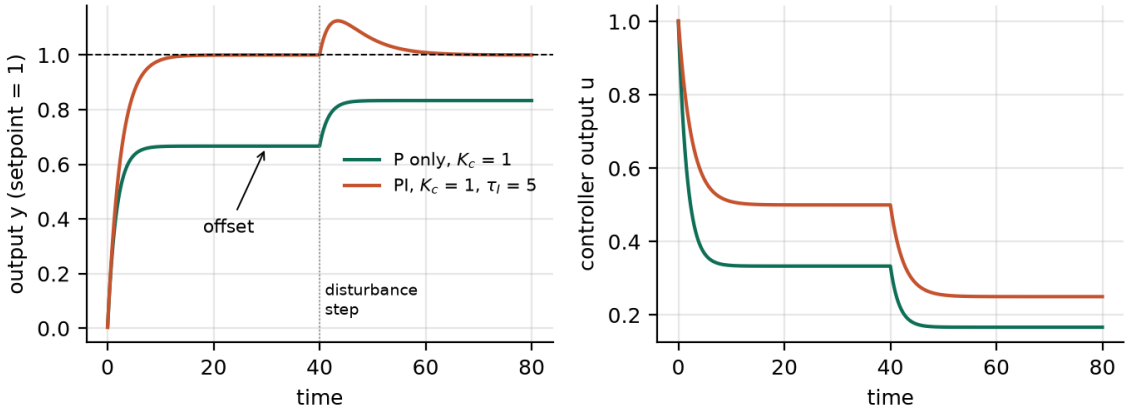


Figure 4: Process $2/(5s + 1)$. Setpoint step at $t = 0$, disturbance step at $t = 40$. P control leaves an offset, which the disturbance makes larger; PI returns exactly to the setpoint both times.

And D? A derivative term $K_c\tau_D\,\mathrm{d}e/\mathrm{d}t$ (the D in PID) anticipates where the error is heading. It helps slow processes with little noise, such as temperature. On noisy biological signals it amplifies noise, so it is often left out, as it is in the rocker.

12 Too much gain: instability

Take three first-order lags in series, $G_p = 1/(s + 1)^3$, under P control. The characteristic equation is

$$1 + \frac{K_c}{(s+1)^3} = 0 \iff s^3 + 3s^2 + 3s + 1 + K_c = 0.$$

At the boundary of stability a pair of poles sits on the imaginary axis, $s = i\omega$. Substitute and split into real and imaginary parts:

$$\text{imaginary: } -\omega^3 + 3\omega = 0 \Rightarrow \omega = \sqrt{3}, \qquad \text{real: } -3\omega^2 + 1 + K_c = 0 \Rightarrow K_c = 8.$$

So the loop is stable for $K_c < 8$, oscillates forever at $K_c = 8$ with period $2\pi/\sqrt{3} \approx 3.6$, and grows without limit above 8. $K_u = 8$ is the **ultimate gain** and $P_u = 3.6$ the **ultimate period**.

> Every lag and delay in the loop makes the controller's information older. A controller that pushes hard on old information overshoots, then over-corrects: the shower again. Lags and delays limit how aggressive a feedback loop can be.

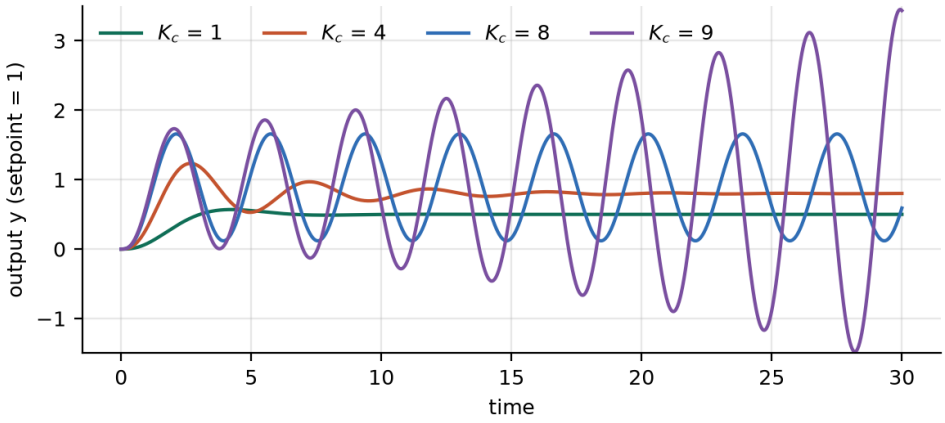


Figure 5: P control of $1/(s+1)^3$. Higher gain: faster, less offset, more oscillation, then instability.

13 Tuning: choosing K_c and τ_I

13.1 Model-based (IMC, or lambda) tuning

Choose how fast you want the closed loop to be, λ, and solve for the controller. For a first-order process, set $\tau_I = \tau$ so the controller's zero cancels the process pole:

$$G_c G_p = K_c \frac{\tau s + 1}{\tau s} \cdot \frac{K}{\tau s + 1} = \frac{K K_c}{\tau s} \quad \Longrightarrow \quad \frac{Y}{R} = \frac{1}{\lambda s + 1}, \quad \lambda = \frac{\tau}{K K_c}.$$

$$\boxed{\tau_I = \tau, \qquad K_c = \frac{\tau}{K\lambda}} \tag{10}$$

The closed loop is first order with time constant λ: no overshoot, no offset. The PI curve in figure 4 is exactly this ($\lambda = 5/(2 \cdot 1) = 2.5$). For a process with dead time θ (FOPDT), a widely used version (Skogestad's SIMC rules) is

$$K_c = \frac{\tau}{K(\lambda + \theta)}, \qquad \tau_I = \min\big(\tau,\ 4(\lambda + \theta)\big),$$

with $\lambda = \theta$ as a good default. Smaller λ: faster but less robust to model error.

13.2 Experimental (Ziegler–Nichols) tuning

With no model, raise K_c on a P-only loop until it oscillates steadily; record K_u and P_u. Then for PI: $K_c = 0.45\,K_u$, $\tau_I = P_u/1.2$. For the example above, $K_c = 3.6$ and $\tau_I = 3.0$. This gives aggressive, rather oscillatory tuning and pushes a real process to the edge of instability during the test, so it is mainly of historical and teaching interest.

13.3 What each knob does

Change	Speed of response	Overshoot / oscillation	Offset
Increase K_c	faster	more	smaller (P) / none (PI)
Decrease τ_I (more integral)	faster return to setpoint	more	none
Add derivative τ_D	slightly faster	less	no effect

Part III
Practice, and the rocker

14 Direct and reverse action

If increasing u increases y (the process gain K is positive), a positive error should increase u: **direct action**. If increasing u *decreases* y ($K < 0$), the sign must be flipped: **reverse action**. Get it wrong and the controller pushes the wrong way, which is guaranteed instability. In the rocker, the pH loop is reverse acting (more CO_2, lower pH); the code sets `reverse=True`, which uses $e = y - r$.

15 Saturation and integral windup

Real actuators have limits. When the controller asks for more than the actuator can give, the actuator stays at its limit while the error persists, and the integral keeps growing ("winding up"). When the error finally reverses, the controller must first unwind that excess integral before the output leaves the limit. Result: a large overshoot.

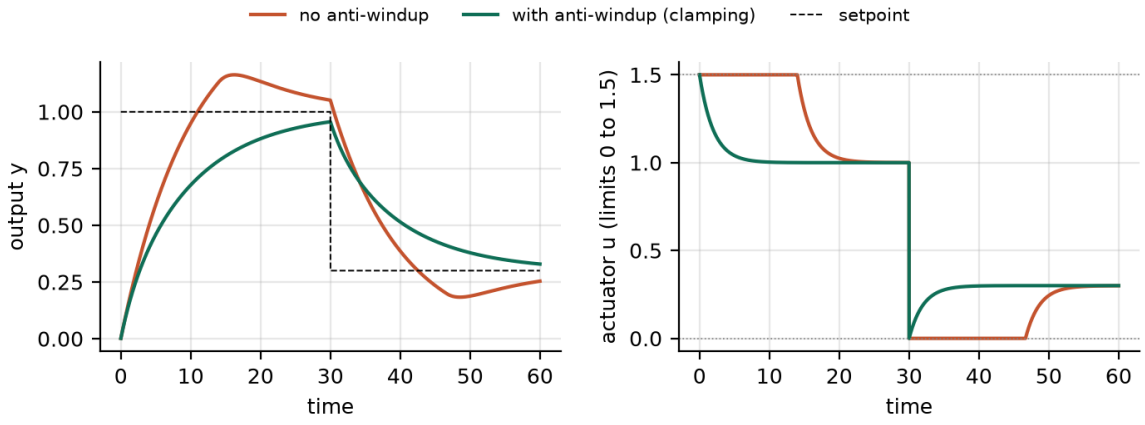


Figure 6: PI control with an actuator limited to $[0, 1.5]$. Without anti-windup (orange) the actuator stays saturated long after it should have backed off, and the output overshoots in both directions.

Anti-windup by clamping, as in `control/pid.py`: compute the unlimited output $\tilde{u}$, limit it to get u, then remove the excess from the integral:

$$I \leftarrow I - (\tilde{u} - u).$$

The integral then always sits exactly where the output just touches its limit, and the controller responds immediately when the error changes sign. In the rocker this matters for the DO loop during the first five days, when DO is above setpoint and the rock rate is already at its minimum.

16 Digital control

A computer controller does not act continuously: it reads the sensor every Δt, computes, and holds its output until the next step (a **zero-order hold**). The integral becomes a sum:

$$e_k = r - y_k, \qquad I_k = I_{k-1} + K_c \frac{\Delta t}{\tau_I} e_k, \qquad u_k = \mathrm{clip}\big(u_{\mathrm{bias}} + K_c e_k + I_k\big). \tag{11}$$

This behaves like the continuous controller as long as Δt is much shorter than the closed-loop time constant; a common rule is $\Delta t \le \lambda/10$. The rocker uses $\Delta t = 5$ min against closed-loop times of about an hour. The

bias u_{bias} is the output when the error and integral are zero, a sensible starting point (5% CO_2 for the pH loop).

17 Beyond one loop

- **Split range**: one controller output drives two actuators in sequence. The rocker's DO controller output 0–50% raises the rock rate from 6 to 25 rpm; 50–100% keeps 25 rpm and raises the O_2 fraction from 21% to 60%.
- **Cascade**: an outer, slow controller sets the setpoint of an inner, fast one. The inner loop rejects fast disturbances before the outer loop even sees them. The project's architecture is a cascade in spirit: the rocker holds pH and DO (minutes); a supervisory layer will choose setpoints and harvest time (hours to days).
- **Feedforward**: measure a disturbance and act *before* it moves y. Example: increase the rock rate as the cell count rises, instead of waiting for DO to fall. Feedback then cleans up the remainder.
- **Schedules**: some inputs are not feedback controlled at all. Perfusion in the rocker follows a fixed ramp (off until day 3, then up to 1.5 vessel volumes per day by day 5).

18 Putting it together: the two loops in the rocker

18.1 The pH loop: a pure-gain process

In the model, pH follows Henderson–Hasselbalch:

$$\text{pH} = 6.1 + \log_{10} \frac{B}{s\,\kappa\,y_{CO_2}}.$$

pH reacts to the CO_2 fraction *instantly*, with no lag, so the process is a pure gain. Bicarbonate B falls slowly as lactic acid accumulates; that is the disturbance. Linearise in y_{CO_2}:

$$K = \frac{\partial\,\text{pH}}{\partial y_{CO_2}} = -\frac{1}{\ln 10\; y_{CO_2}} \approx -5.9 \quad \text{at } y_{CO_2} = 0.073.$$

The gain is negative, hence reverse action. With reverse action the loop sees $|K|$, so take $G_p = |K|$ and the PI controller (9). The closed loop is

$$\frac{Y}{R} = \frac{G_c|K|}{1 + G_c|K|} = \frac{K_c|K|\,(\tau_I s + 1)}{(1 + K_c|K|)\,\tau_I s + K_c|K|}.$$

One pole, at $s = -K_c|K|/\big((1 + K_c|K|)\tau_I\big)$, so the closed-loop time constant is

$$T_{\text{cl}} = \tau_I\,\frac{1 + K_c|K|}{K_c|K|}.$$

The initial value theorem gives the immediate jump after a setpoint step: $\lim_{s\to\infty} Y/R = K_c|K|/(1 + K_c|K|)$. With the rocker's $K_c = 0.04$ and $\tau_I = 0.25$ h, $K_c|K| = 0.24$:

- an immediate jump of $0.24/1.24 \approx 19\%$ of the step (the proportional part acting on a static process);
- then a first-order approach with $T_{\text{cl}} = 0.25 \times 1.24/0.24 \approx 1.3$ h.

The simulation agrees: after a step from pH 7.20 to 7.10 on day 7, the response covers 63% of the step in 1 h, 82% in 2 h and 92% in 3 h. Note that the pH loop could be made faster by raising K_c; the process itself has no lag, so the limits come from the 5-minute sampling (which acts like a small delay) and from sensor noise, which a higher gain amplifies.

18.2 The DO loop: a first-order process

The oxygen balance is first order with $\tau = 1/k_L a$, about 0.5 h at 15 rpm. Around 50% DO and 15 rpm, a one-rpm increase raises steady-state DO by about 2.7 percentage points, and one percent of controller output moves the rocker by 0.38 rpm, so the process gain seen by the controller is about 1 (% DO per % output). The rocker uses $K_c = 1.5$ and $\tau_I = 1$ h. IMC would suggest $\tau_I = \tau \approx 0.5$ h and gives $\lambda = \tau/(KK_c) \approx 0.3$ h; the longer τ_I is a little more conservative, which is sensible because τ and K change as the rock rate changes. The DO step in the technical document settles within a couple of hours without overshoot.

18.3 What feedback cannot fix

The controller controls the measurement, not the truth

If the pH sensor drifts by -0.005 per day, the loop keeps the *reading* at 7.20, so the true pH rises by 0.005 per day. Feedback is blind to sensor bias. Only an independent measurement or a model can catch it, which is where state estimation comes in.

19 Bridge to state estimation

Everything above used transfer functions, which describe input–output behaviour. The same systems can be written in **state-space** form, which is the language you know from estimation:

$$\dot{x} = Ax + Bu, \qquad y = Cx.$$

Taking Laplace transforms gives the link:

$$G(s) = C\,(sI - A)^{-1}B,$$

and the poles of $G(s)$ are the **eigenvalues of** A. "All poles in the left half-plane" and "all eigenvalues of A have negative real parts" are the same statement.

- A state-feedback controller $u = -Fx$ moves the eigenvalues of $A - BF$; an observer with gain L moves those of $A - LC$. Placing controller poles and placing observer poles are mirror images of each other (**duality**): the Kalman filter is the dual of the linear-quadratic regulator.
- For linear systems, the **separation principle** says you can design the estimator and the controller independently and then connect them: control the *estimated* state as if it were the true one.
- That is the plan for this project: the rocker's PI loops stay as they are, and an ensemble Kalman filter on top estimates the hidden states (memory share, patient parameters) that the supervisory layer will act on.

Exercises

1. A first-order process reaches 63% of its final change 4 minutes after a step, and settles 3 units higher after an input step of 2 units. What are K and τ?

2. Write the transfer function from inflow to level for a tank with $A = 2$ m^2 and $R = 3$ min/m^2.

3. Use the final value theorem to find where $Y(s) = \dfrac{4}{s(2s+1)(s+3)}$ ends up.

4. A process $2/(10s + 1)$ is under P control with $K_c = 4$. Find the closed-loop time constant and the steady-state offset after a unit setpoint step.

5. Design a PI controller for the process in exercise 4 so the closed loop has time constant 2.

6. Is $G(s) = 1/(s^2 - s + 1)$ stable? What would its step response look like?

Answers

1. $K = 3/2 = 1.5$, $\tau = 4$ min.

2. $\tau s H + H = RQ$, so $G(s) = R/(AR\,s + 1) = 3/(6s + 1)$: gain 3 min/m^2, time constant 6 min.

3. $\lim_{s\to 0} sY(s) = 4/(1 \cdot 3) = 4/3$.

4. $KK_c = 8$: $\tau_{\text{cl}} = 10/9 \approx 1.1$; $y(\infty) = 8/9$, so the offset is $1/9 \approx 11\%$.

5. $\tau_I = \tau = 10$, $K_c = \tau/(K\lambda) = 10/(2 \cdot 2) = 2.5$.

6. The poles are $s = (1 \pm i\sqrt{3})/2$: positive real part, so unstable. The response oscillates with a growing amplitude.